Chapitre 4 : La régression

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Table des matières

1 Avant de commencer

Ce chapitre reprends les codes du chapitre 4 sur la régression du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2ème édition).

```
library(tidyverse)
```

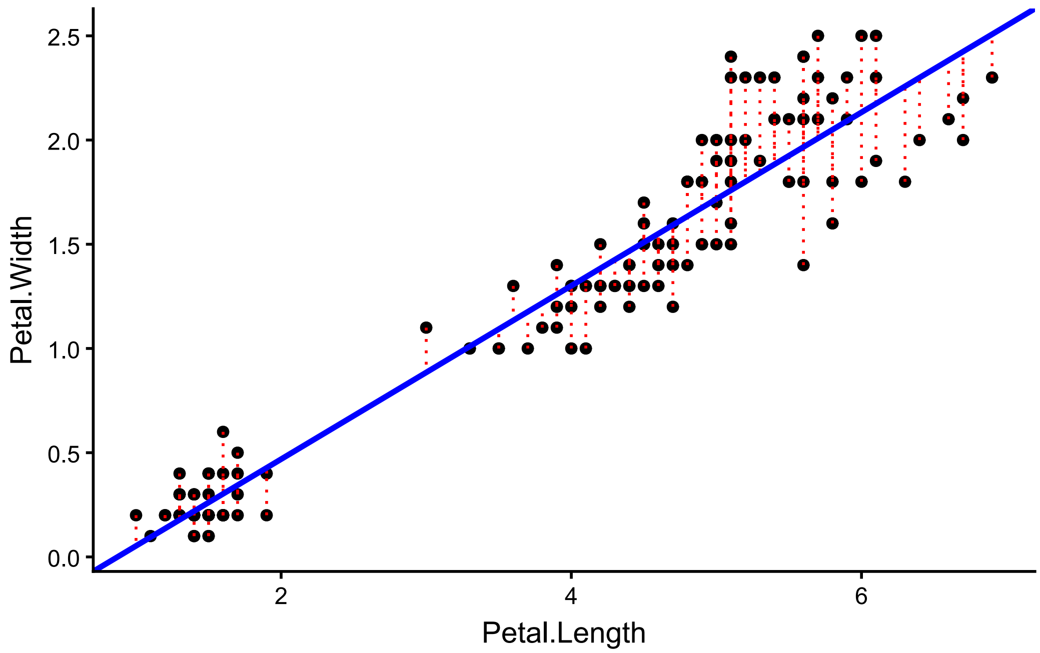
```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.0     ✔ readr     2.1.6
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.2     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

2 Régression linéaire simple

2.1 graphique pour visualiser les moindres carrés

```
ggplot(iris) +
  aes(
    x = Petal.Length,
    y = Petal.Width
  ) +
  geom_point() +
  geom_segment(
    aes(
      x = Petal.Length,
      xend = Petal.Length,
      y = Petal.Width,
      yend = 0.416 * Petal.Length - 0.363
    ),
    linetype = "dotted",
    color = "red"
  ) +
  geom_abline(slope = 0.416, intercept = -0.363, color = "blue", size = 1) +
  theme_classic()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



2.2 modèle linéaire regression simple

```r
regression_lineaire <- lm(
  formula = Petal.Width ~ Petal.Length,
  data = iris
  )

summary(regression_lineaire)
```

```


Call:
lm(formula = Petal.Width ~ Petal.Length, data = iris)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.56515 -0.12358 -0.01898  0.13288  0.64272 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -0.363076   0.039762  -9.131  4.7e-16 ***
```

```
Petal.Length  0.415755   0.009582  43.387   < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2065 on 148 degrees of freedom
Multiple R-squared:  0.9271,    Adjusted R-squared:  0.9266
F-statistic:  1882 on 1 and 148 DF,  p-value: < 2.2e-16
```

2.3 représentation graphique pour données autre que linéaires

```
purrr::map2(
  .x = c("x1", "x2", "x3", "x4"),
  .y = c("y1", "y2", "y3", "y4"),
  .f = ~ ggplot(anscombe) +
    aes(x = get(.x), y = get(.y)) +
    geom_point() +
    geom_abline(slope = 0.5, intercept = 3, color = "blue") +
    annotate(
      "text",
      x = 14,
      y = 3,
      label = "y ~ 0,5 * x + 3 \n R² = 0.63 \n p-valeur < 0.01"
    ) +
    labs(x = .x, y = .y) +
    coord_cartesian(xlim = c(5, 19), ylim = c(0,13)) +
    theme_classic()
)|>
  purrr::map2(
       .y  =  c("img/04Rl03N.png",  "img/04Rl03bisN.png",  "img/04Rl03terN.png",
"img/04Rl03quaN.png"),
    .f = ~ ggsave(plot = .x, filename = .y, width = 6, height = 6, units = "cm")
  )
```

```
[[1]]
[1] "img/04Rl03N.png"

[[2]]
[1] "img/04Rl03bisN.png"

[[3]]
[1] "img/04Rl03terN.png"
```

```
[[4]]
[1] "img/04Rl03quaN.png"
```

```
purrr::map2(
  .x = c("x1", "x2", "x3", "x4"),
  .y = c("y1", "y2", "y3", "y4"),
  .f = ~ lm(get(.y) ~ get(.x), data = anscombe) |> summary()
)
```

```
[[1]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
     Min       1Q   Median       3Q      Max
-1.92127 -0.45577 -0.04136  0.70941  1.83882

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0001     1.1247   2.667  0.02573 *
get(.x)       0.5001     0.1179   4.241  0.00217 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6665,    Adjusted R-squared:  0.6295
F-statistic: 17.99 on 1 and 9 DF,  p-value: 0.00217


[[2]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
    Min      1Q  Median      3Q     Max
-1.9009 -0.7609  0.1291  0.9491  1.2691

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.001      1.125   2.667  0.02576 *
```

```
get(.x)        0.500      0.118   4.239  0.00218 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6662,    Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002179


[[3]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
    Min      1Q  Median      3Q     Max
-1.1586 -0.6146 -0.2303  0.1540  3.2411

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0025     1.1245   2.670  0.02562 *
get(.x)       0.4997     0.1179   4.239  0.00218 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6663,    Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002176


[[4]]

Call:
lm(formula = get(.y) ~ get(.x), data = anscombe)

Residuals:
   Min     1Q Median     3Q    Max
-1.751 -0.831  0.000  0.809  1.839

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0017     1.1239   2.671  0.02559 *
get(.x)       0.4999     0.1178   4.243  0.00216 **
```

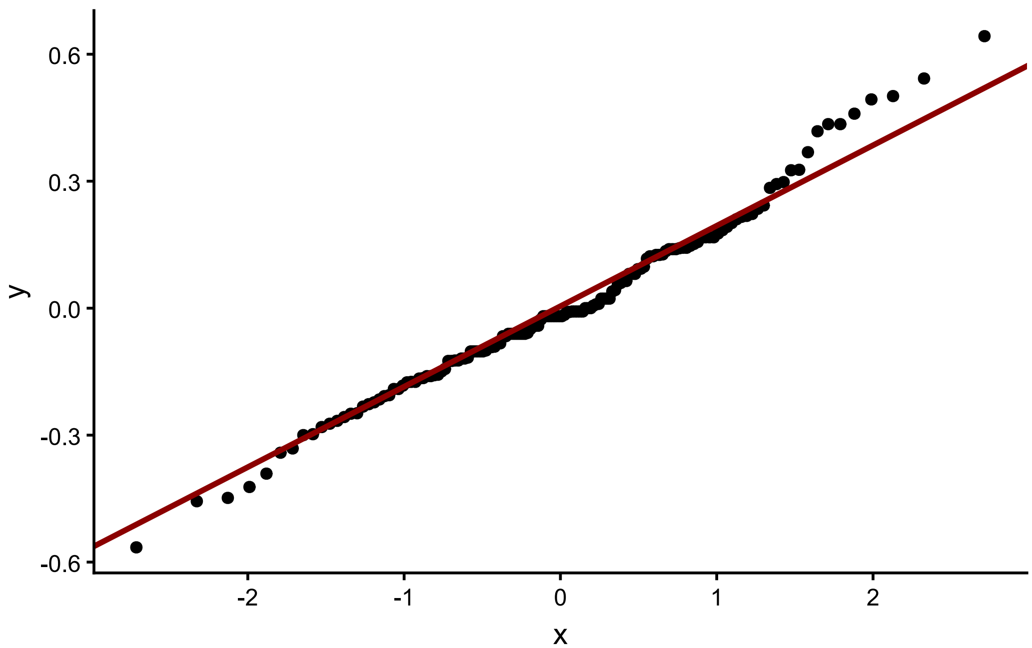
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6667,    Adjusted R-squared:  0.6297
F-statistic:    18 on 1 and 9 DF,  p-value: 0.002165
```

2.4 Normalité des résidus

```
shapiro.test(regression_lineaire$residuals)
```

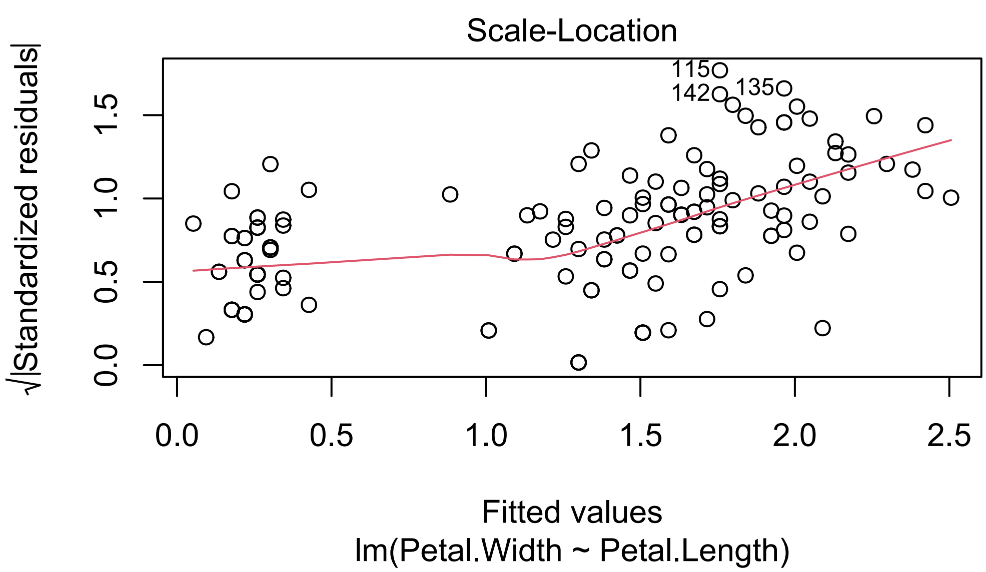
```


    Shapiro-Wilk normality test

data:  regression_lineaire$residuals
W = 0.98378, p-value = 0.07504
```

```
tibble(
  residus = regression_lineaire$residuals
) |>
  ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", size = 1) +
  theme_classic()
```



2.5 homogéinité

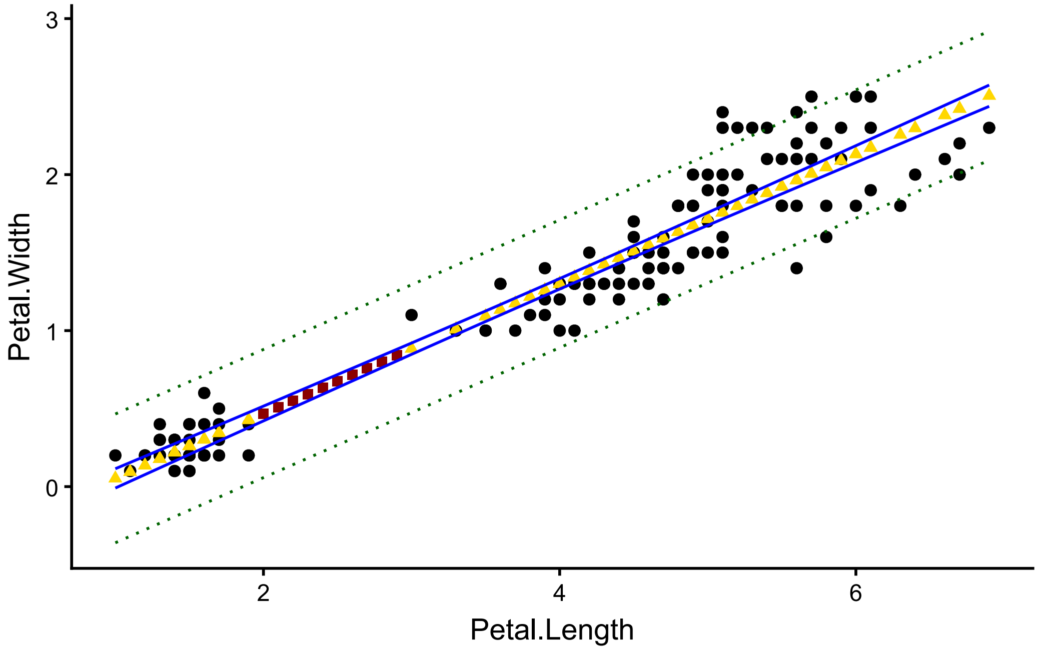
```
plot(regression_lineaire, which = 3)
```



2.6 intervalle de confiance

```
iris|>
  ggplot() +
  aes(
    x = Petal.Length,
    y = Petal.Width
    ) +
  geom_point() +
  geom_point(aes(y = predict(regression_lineaire)), shape = 17, color = "gold") +
  geom_point(
    data = tibble(
      Petal.Length = seq(2, 2.9, 0.1),
    Petal.Width = predict(regression_lineaire, newdata = data.frame(Petal.Length
= seq(2, 2.9, 0.1)))
    ),
    color = "dark red",
    shape = 15
    ) +
  geom_line(aes(y = predict(regression_lineaire, interval = c("confidence"))[,
2]), color = "blue") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("confidence"))[,
3]), color = "blue") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("prediction"))[,
2]), color = "darkgreen", linetype = "dotted") +
  geom_line(aes(y = predict(regression_lineaire, interval = c("prediction"))[,
3]), color = "darkgreen", linetype = "dotted") +
  theme_classic()
```

```
Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
Warning in predict.lm(regression_lineaire, interval = c("prediction")): les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
```



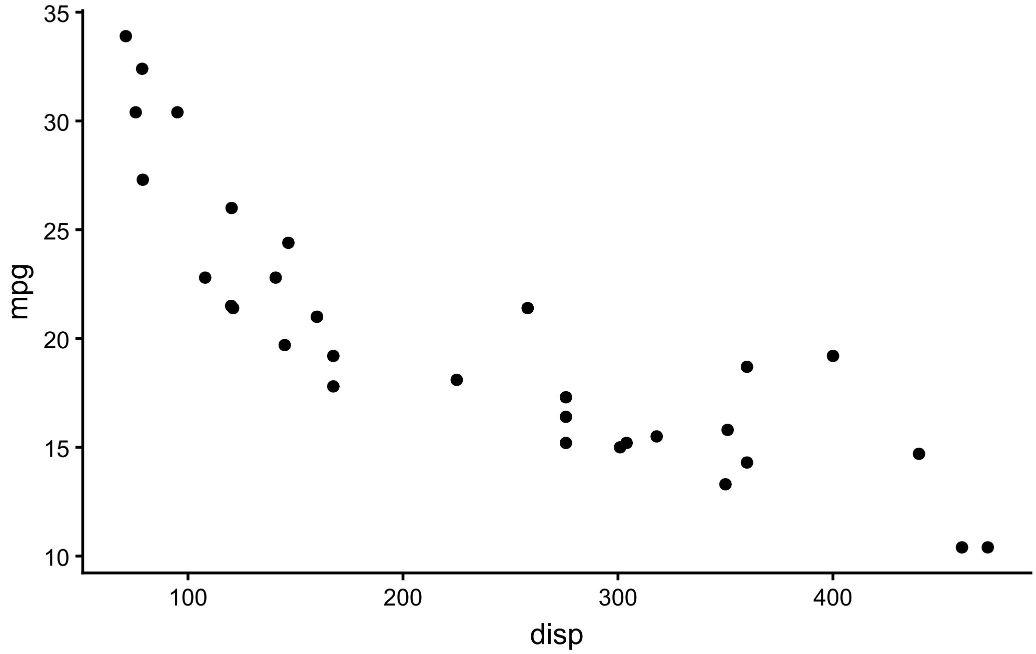

```
ggsave("img/04Rl07N.png", width = 14, height = 7, units = "cm")
```

```
Warning  in  predict.lm(regression_lineaire,  interval  =  c("prediction")):  les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
Warning  in  predict.lm(regression_lineaire,  interval  =  c("prediction")):  les
prédictions sur les données actuelles se réfèrent à des réponses _futures_
```

3 Régression polynomiale

3.1 graphique

```
ggplot(mtcars) +
  aes(y = mpg, x = disp) +
  geom_point() +
  theme_classic()
```



3.2 régression linéaire

```
regression_lineaire_cars <- lm(mpg ~ disp, data = mtcars)
summary(regression_lineaire_cars)
```

```

Call:
lm(formula = mpg ~ disp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.8922 -2.2022 -0.9631  1.6272  7.2305 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 29.599855   1.229720  24.070  < 2e-16 ***
disp        -0.041215   0.004712  -8.747 9.38e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.251 on 30 degrees of freedom
Multiple R-squared:  0.7183,	Adjusted R-squared:  0.709 
F-statistic: 76.51 on 1 and 30 DF,  p-value: 9.38e-10
```

vérification de normalité des résidus

```r
shapiro.test(regression_lineaire_cars$residuals)
```

```

	Shapiro-Wilk normality test

data:  regression_lineaire_cars$residuals
W = 0.9271, p-value = 0.03255
```

3.3 régression polynomiale

```r
regression_polynomiale <- lm(mpg ~ I(disp^2) + disp, data = mtcars)
summary(regression_polynomiale)
```

```

Call:
lm(formula = mpg ~ I(disp^2) + disp, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.9112 -1.5269 -0.3124  1.3489  5.3946 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.583e+01  2.209e+00  16.221 4.39e-16 ***
I(disp^2)    1.255e-04  3.891e-05   3.226   0.0031 ** 
disp        -1.053e-01  2.028e-02  -5.192 1.49e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.837 on 29 degrees of freedom
Multiple R-squared:  0.7927,	Adjusted R-squared:  0.7784 
F-statistic: 55.46 on 2 and 29 DF,  p-value: 1.229e-10
```

vérification normalité des résidus

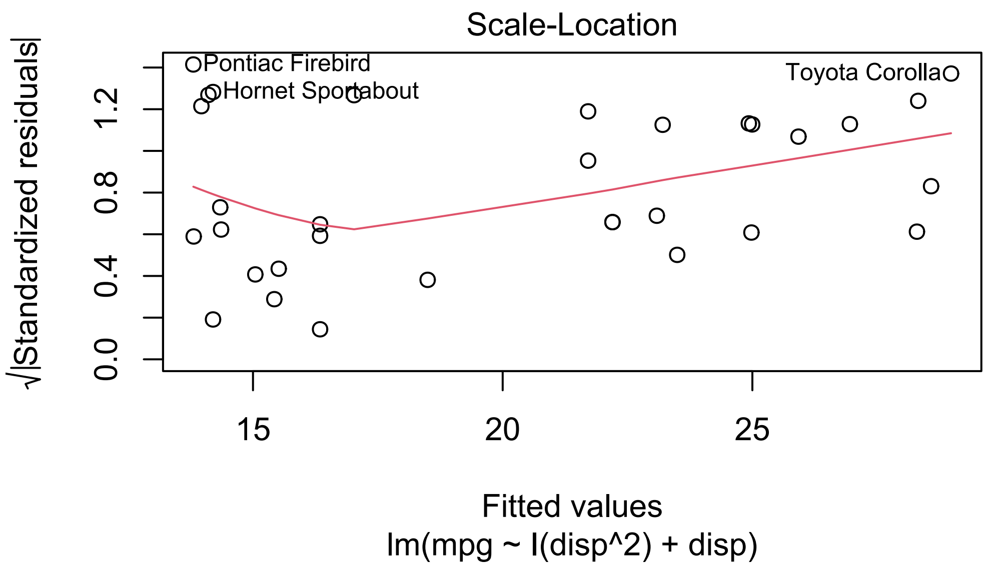
```r
shapiro.test(regression_polynomiale$residuals)
```

```

	Shapiro-Wilk normality test
```

```
data:  regression_polynomiale$residuals
W = 0.93679, p-value = 0.06073
```

vérification homogénéité

```
plot(regression_polynomiale, which = 3)
```



3.4 comparaison des deux modèles

```
AIC(regression_lineaire_cars, regression_polynomiale)
```

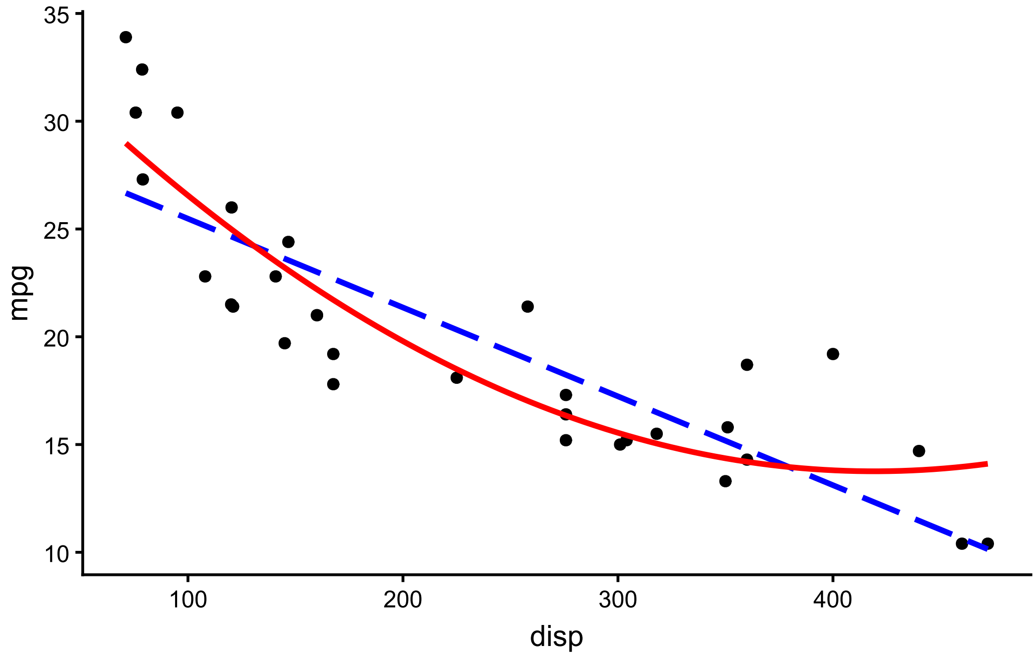
```
                         df      AIC
regression_lineaire_cars  3 170.2094
regression_polynomiale    4 162.3957
```

```
anova(regression_lineaire_cars, regression_polynomiale)
```

```
Analysis of Variance Table

Model 1: mpg ~ disp
Model 2: mpg ~ I(disp^2) + disp
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
```

```
1     30 317.16
2     29 233.39  1    83.766 10.408 0.003104 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ggplot(mtcars) +
  aes(y = mpg, x = disp) +
  geom_point() +
  geom_smooth(method = lm, formula = y ~ x, colour="blue", se = FALSE, linetype
= "longdash") +
  geom_smooth(method = lm, formula = y ~ x + I(x^2), color = "red", se = FALSE) +
  theme_classic()
```




```
ggsave("img/04Rl16N.png", width = 14, height = 7, units = "cm")
```

4 Régression exponentielle

4.1 graphique

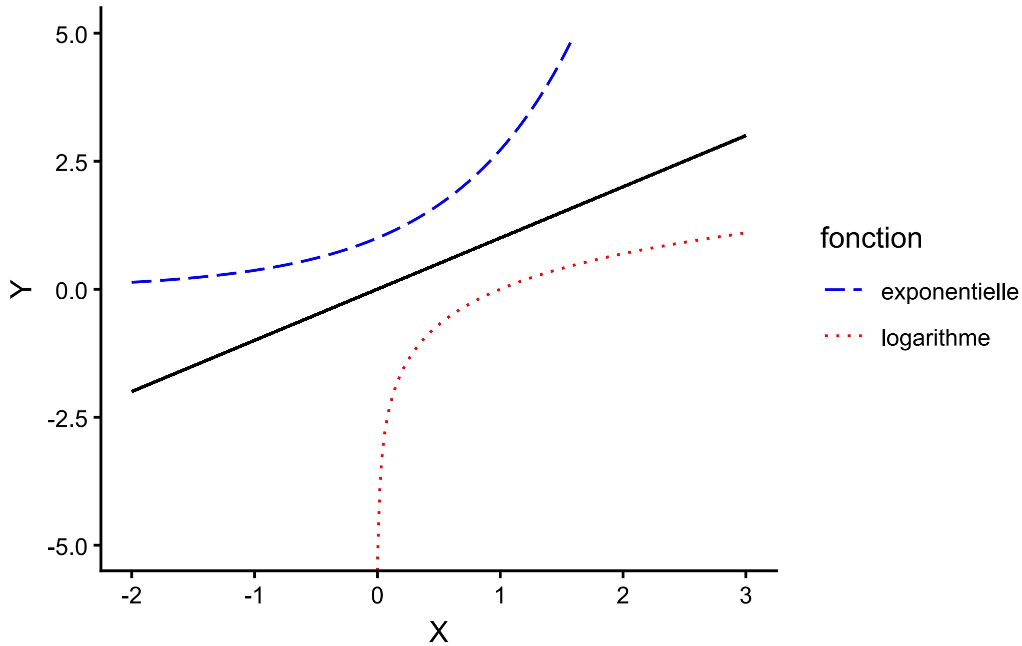
```
intervalle <- seq(-2.5, 3.5, 0.01)

tibble(
  X = rep(intervalle, 2),
```

```r
  Y = c(exp(intervalle), log(intervalle)),
  fonction = rep(c("exponentielle", "logarithme"), each = length(intervalle))
)|>
  ggplot() +
  aes(x = X,
      y = Y,
      color = fonction,
      linetype = fonction
      ) +
  geom_line() +
  geom_segment(
    aes(x = -2, xend = 3, y = -2, yend = 3),
    color = "black",
    linetype = "solid"
    ) +
  xlim(-2, 3) +
  ylim(-5, 5) +
  scale_colour_manual(values = c("blue", "red")) +
  scale_linetype_manual(values = c("longdash", "dotted")) +
  theme_classic()
```

```
Warning in log(intervalle): Production de NaN
```

```
Warning in geom_segment(aes(x = -2, xend = 3, y = -2, yend = 3), color = "black", :
All aesthetics have length 1, but the data has 1202 rows.
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

```
Warning: Removed 540 rows containing missing values or values outside the scale
range
(`geom_line()`).
```




```
ggsave("img/04Rl17N.png", width = 14, height = 7, units = "cm")
```

```
Warning in geom_segment(aes(x = -2, xend = 3, y = -2, yend = 3), color = "black", :
All aesthetics have length 1, but the data has 1202 rows.
ℹ Please consider using `annotate()` or provide this layer with data containing
  a single row.
Removed 540 rows containing missing values or values outside the scale range
(`geom_line()`).
```

4.2 régression exponentielle

```
regression_exponentielle <-
  lm(
    Ozone ~ exp(Temp),
    data = airquality |>
      filter(Temp < 86)
    )

regression_exponentielle |>
  summary()
```

```


Call:
```

```
lm(formula = Ozone ~ exp(Temp), data = filter(airquality, Temp <
    86))

Residuals:
    Min      1Q  Median      3Q     Max
-40.087 -14.000  -6.643   6.362 139.493

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 2.764e+01  2.828e+00   9.772 1.18e-15 ***
exp(Temp)   5.770e-36  1.533e-36   3.765 0.000302 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25.42 on 87 degrees of freedom
  (30 observations effacées parce que manquantes)
Multiple R-squared:  0.1401,    Adjusted R-squared:  0.1302
F-statistic: 14.17 on 1 and 87 DF,  p-value: 0.0003023
```

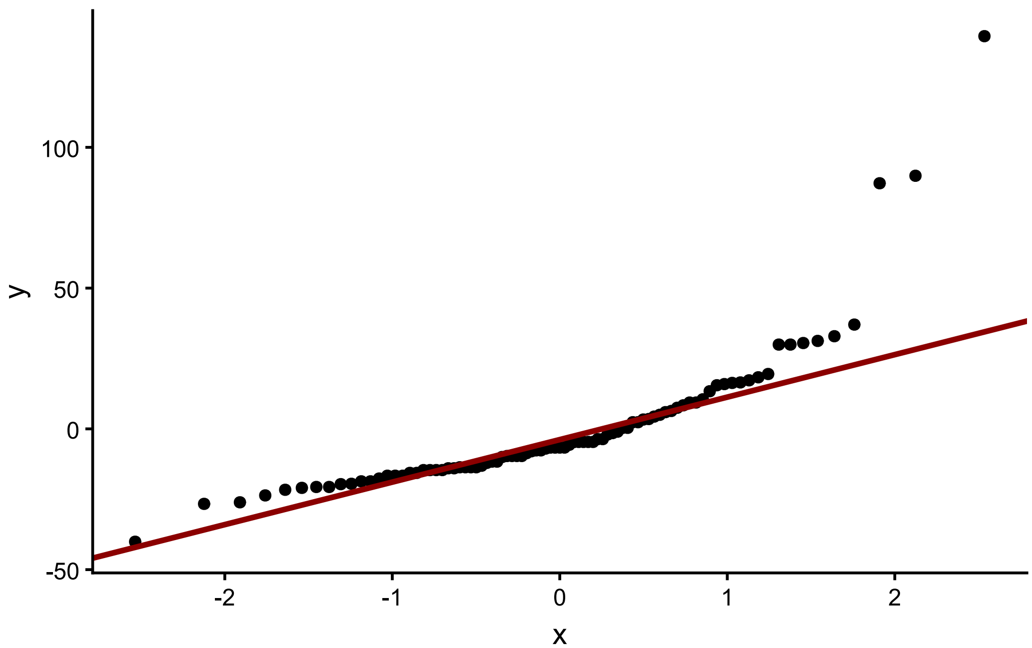
vérification de la normalité

```
shapiro.test(regression_exponentielle$residuals)
```

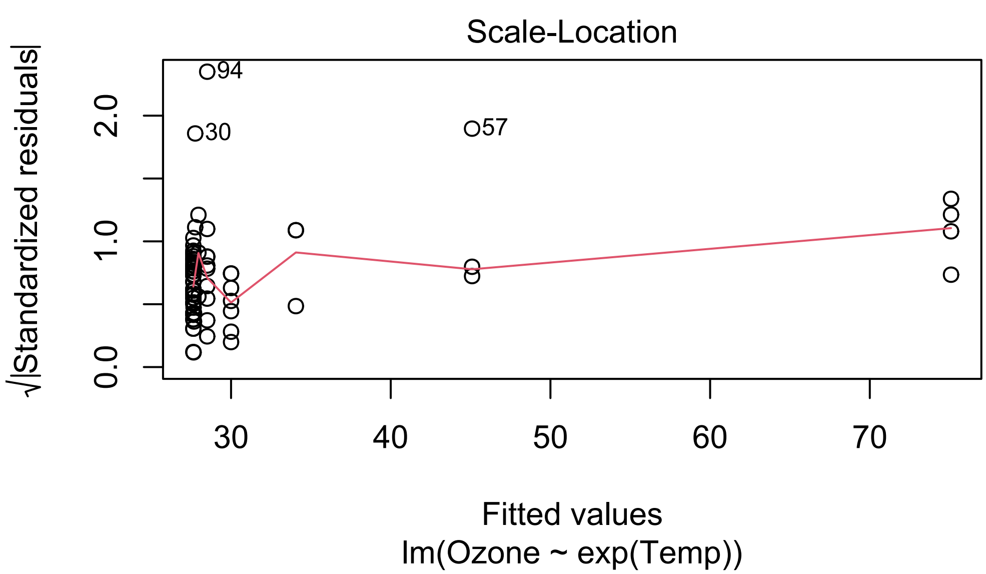
```


    Shapiro-Wilk normality test

data:  regression_exponentielle$residuals
W = 0.73239, p-value = 1.932e-11
```

confirmation par un graphique

```
tibble(
  residus = regression_exponentielle$residuals
) |>
  ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", linewidth = 1) +
  theme_classic()
```




```r
plot(regression_exponentielle, which = 3)
```

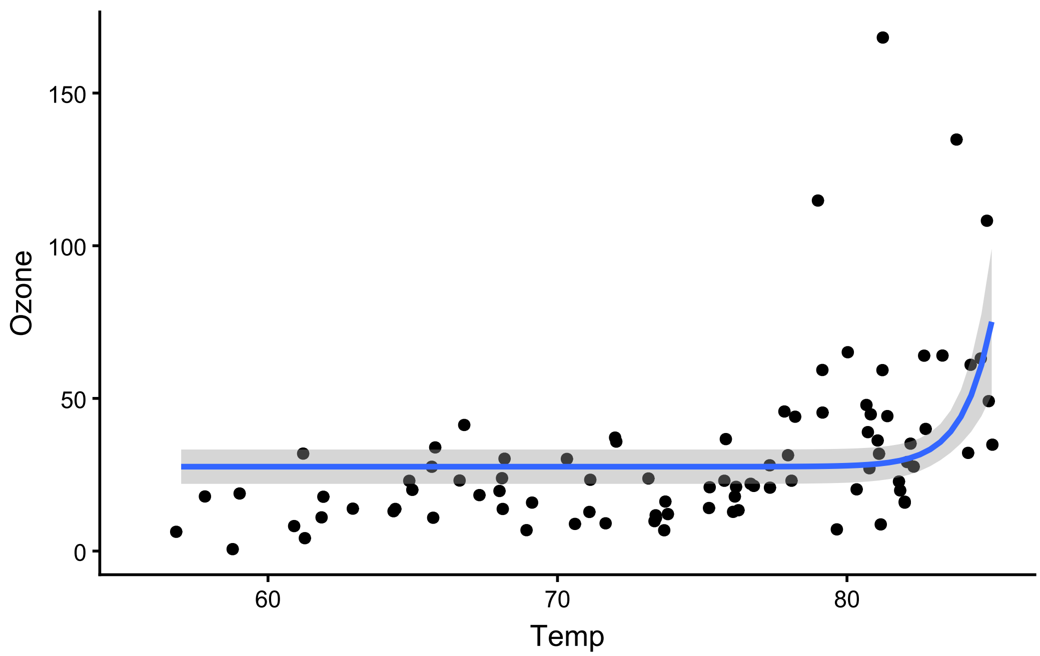



```r
airquality |>
  filter(Temp < 86) |>
```

```r
  ggplot() +
  aes(x = Temp, y = Ozone) +
  geom_jitter() +
  geom_smooth(method = lm, formula = y ~ exp(x)) +
  theme_classic()
```

```
Warning: Removed 30 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 30 rows containing missing values or values outside the scale
range
(`geom_point()`).
```



5 Régression logistique binomiale

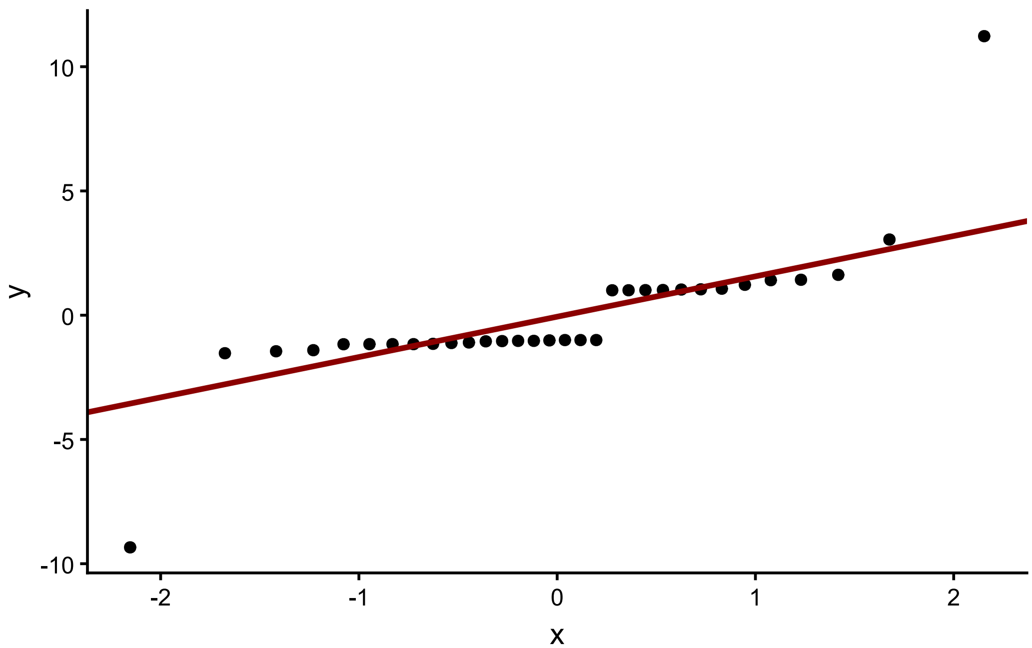
```r
regression_logistique <- glm(
  am ~ wt,
  data = mtcars,
  family = "binomial"
)

summary(regression_logistique)
```

```

Call:
glm(formula = am ~ wt, family = "binomial", data = mtcars)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   12.040      4.510   2.670  0.00759 **
wt            -4.024      1.436  -2.801  0.00509 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 43.230  on 31  degrees of freedom
Residual deviance: 19.176  on 30  degrees of freedom
AIC: 23.176

Number of Fisher Scoring iterations: 6
```

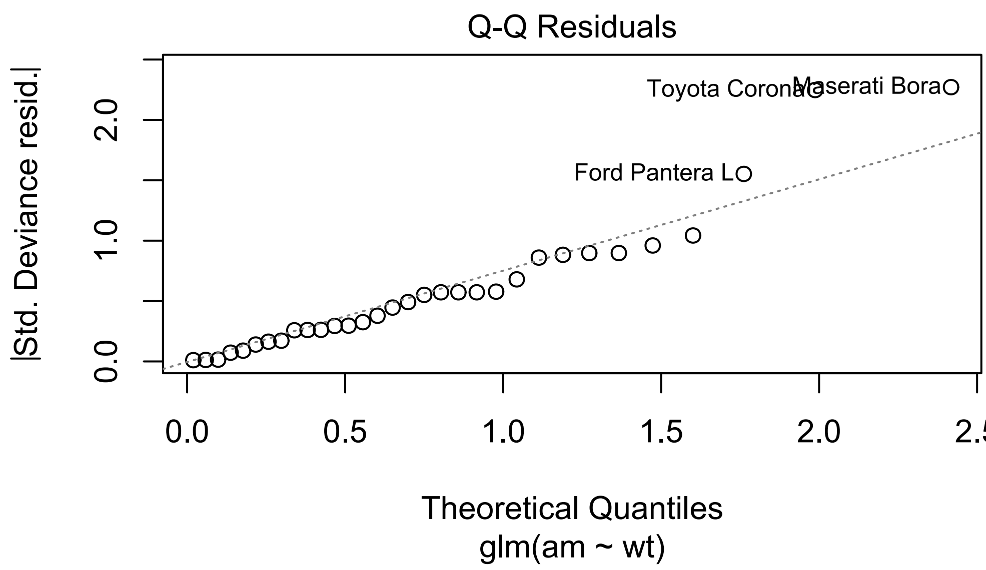
```
shapiro.test(regression_logistique$residuals)
```

```


	Shapiro-Wilk normality test

data:  regression_logistique$residuals
W = 0.69219, p-value = 6.658e-07
```

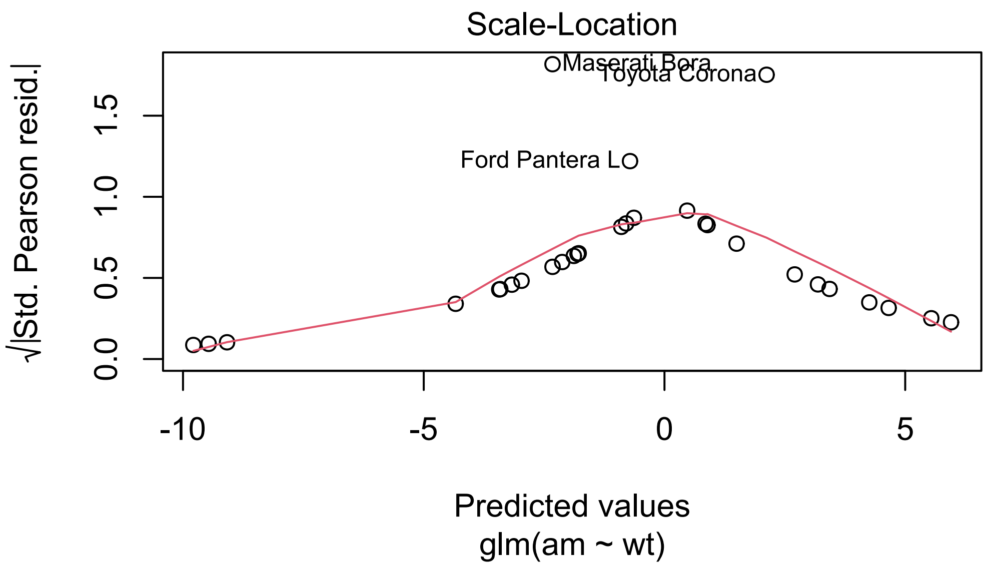
```
tibble(
  residus = regression_logistique$residuals
) |>
  ggplot() +
  aes(sample = residus) +
  geom_qq() +
  geom_qq_line(color = "dark red", size = 1) +
  theme_classic()
```



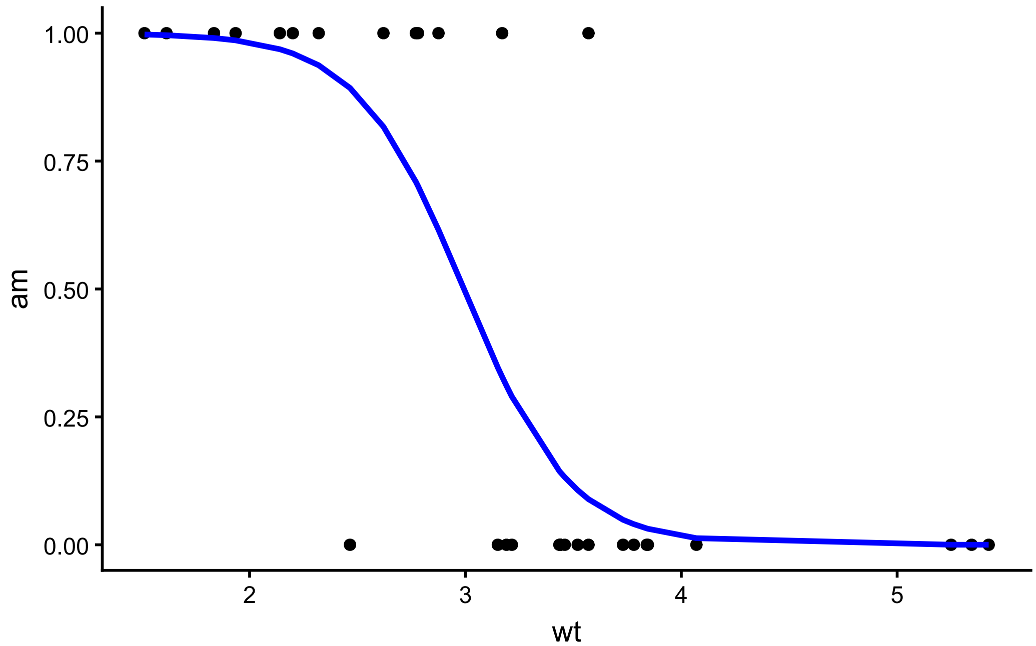

```r
plot(regression_logistique, which = 2)
```




```r
plot(regression_logistique, which = 3)
```

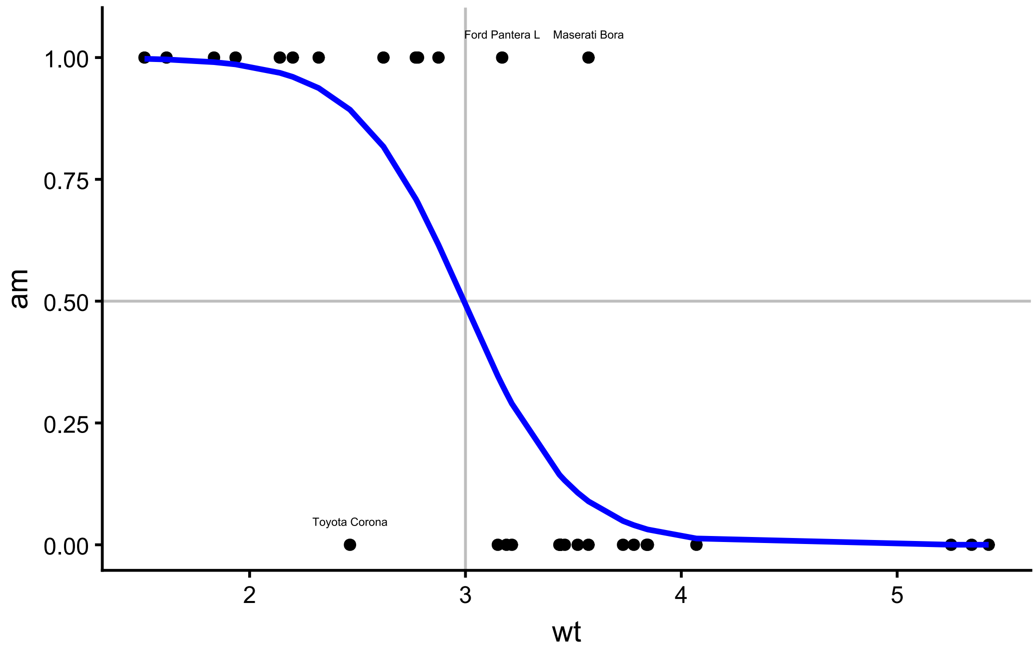



```r
# version simplifiée
mtcars |>
  mutate(
    prediction_am = regression_logistique$fitted.values
    ) |>
  ggplot() +
  aes(x = wt, y = am) +
  geom_point() +
  geom_line(aes(y = prediction_am), color = "blue", linewidth = 1) +
  theme_classic()
```




```
# version annotée
mtcars |>
  rownames_to_column() |>
  mutate(
    prediction_am = regression_logistique$fitted.values,
    etiquette = case_when(
     rowname %in% c("Toyota Corona", "Maserati Bora", "Ford Pantera L") ~ rowname
    )
  ) |>
  ggplot() +
  aes(x = wt, y = am) +
  geom_point() +
  geom_hline(yintercept = 0.5, color = "grey") +
  geom_vline(xintercept = 3, color = "grey") +
  geom_line(aes(y = prediction_am), color = "blue", linewidth = 1) +
  geom_text(aes(label = etiquette), size = 1.5, nudge_y = 0.05) +
  theme_classic()
```

```
Warning: Removed 29 rows containing missing values or values outside the scale
range
(`geom_text()`).
```



6 Régression multiple

6.1 Régression multiple avec espèce

```
regression_multiple_espece <- lm(
  formula = Petal.Width ~ Petal.Length * Species,
  data = iris
  )

regression_multiple_espece |>
  summary()
```

```


Call:
lm(formula = Petal.Width ~ Petal.Length * Species, data = iris)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.6337 -0.0744 -0.0134  0.0866  0.4503 

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)    
(Intercept)                    -0.04822    0.21472  -0.225 0.822627    
Petal.Length                    0.20125    0.14586   1.380 0.169813    
```

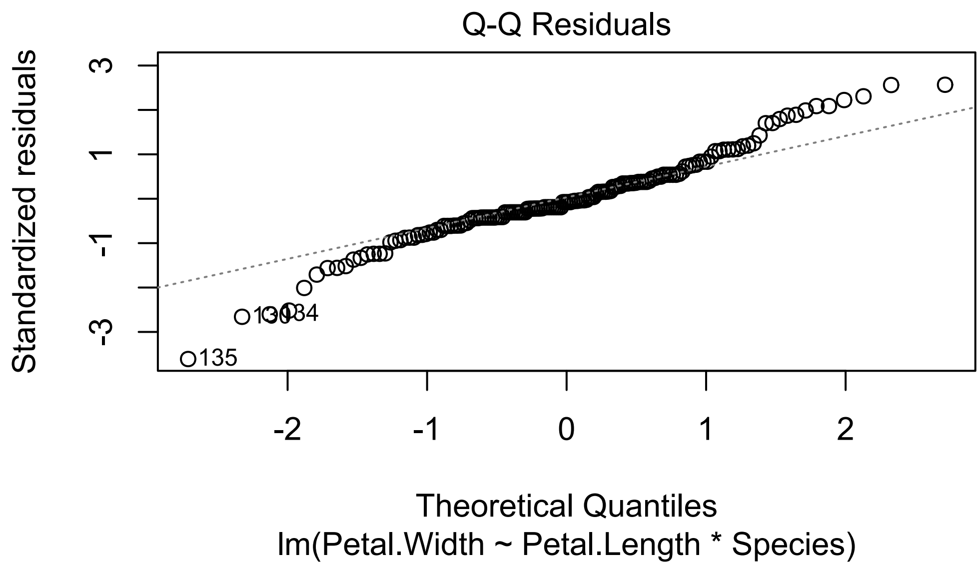
```
Speciesversicolor              -0.03607    0.31538  -0.114 0.909109
Speciesvirginica                1.18425    0.33417   3.544 0.000532 ***
Petal.Length:Speciesversicolor  0.12981    0.15550   0.835 0.405230
Petal.Length:Speciesvirginica  -0.04095    0.15291  -0.268 0.789244
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1773 on 144 degrees of freedom
Multiple R-squared:  0.9477,	Adjusted R-squared:  0.9459
F-statistic: 521.9 on 5 and 144 DF,  p-value: < 2.2e-16
```

```
car::Anova(regression_multiple_espece, type = "3")
```

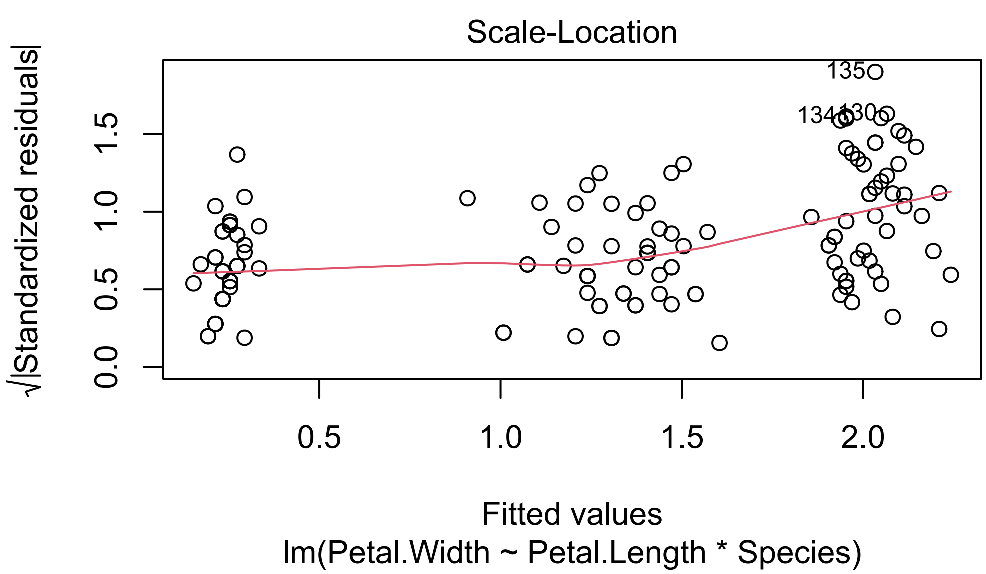
```
Anova Table (Type III tests)

Response: Petal.Width
                     Sum Sq  Df F value    Pr(>F)
(Intercept)          0.0016   1  0.0504 0.8226271
Petal.Length         0.0599   1  1.9036 0.1698135
Species              0.5026   2  7.9931 0.0005107 ***
Petal.Length:Species 0.1842   2  2.9297 0.0566046 .
Residuals            4.5274 144
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

vérification du modèle avec espèce

```
plot(regression_multiple_espece, which = 2)
```




```r
plot(regression_multiple_espece, which = 3)
```



test post-hoc

```r
library(multcomp)
```

```
Le chargement a nécessité le package : mvtnorm
```

```
Le chargement a nécessité le package : survival
```

```
Le chargement a nécessité le package : TH.data
```

```
Le chargement a nécessité le package : MASS
```

```


Attachement du package : 'MASS'
```

```
L'objet suivant est masqué depuis 'package:dplyr':

    select
```

```


Attachement du package : 'TH.data'
```

```
L'objet suivant est masqué depuis 'package:MASS':

    geyser
```

```
summary(
  glht(
    regression_multiple_espece,
    linfct = mcp(
      Species = "Tukey"
      )
    )
  )
```

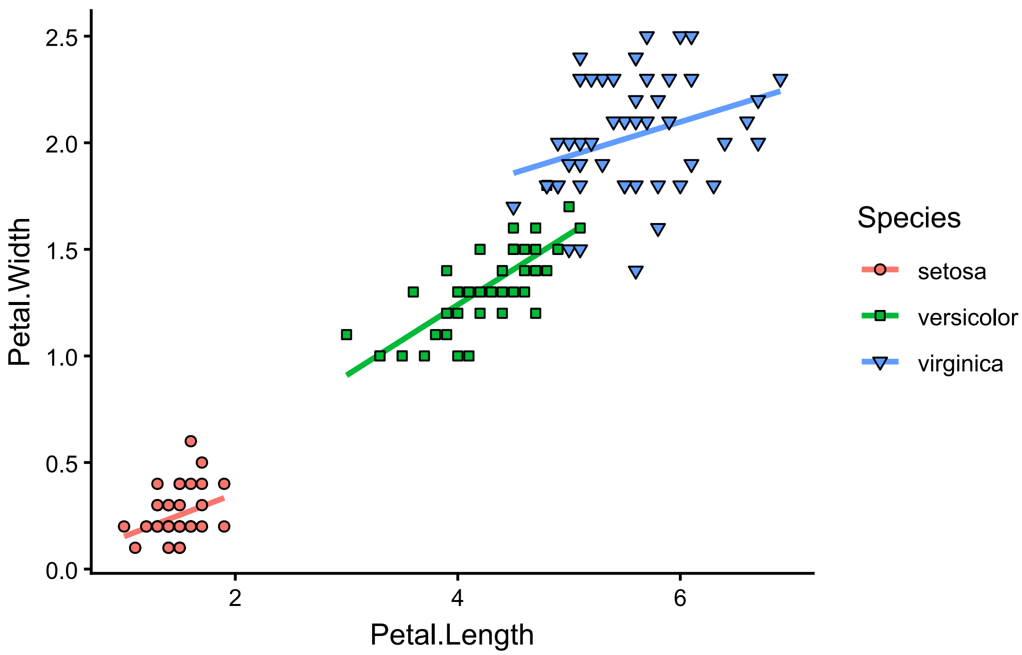
```
Warning in mcp2matrix(model, linfct = linfct): covariate interactions found --
default contrast might be inappropriate
```

```

     Simultaneous Tests for General Linear Hypotheses

```

```
Multiple Comparisons of Means: Tukey Contrasts


Fit: lm(formula = Petal.Width ~ Petal.Length * Species, data = iris)

Linear Hypotheses:
                            Estimate Std. Error t value Pr(>|t|)
versicolor - setosa == 0    -0.03607    0.31538  -0.114  0.99281
virginica - setosa == 0      1.18425    0.33417   3.544  0.00150 **
virginica - versicolor == 0  1.22032    0.34486   3.539  0.00158 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```

```
ggplot(iris) +
  aes(
    x = Petal.Length,
    y = Petal.Width,
    color = Species,
    fill = Species,
    shape = Species
    ) +
  geom_smooth(method = lm, se = FALSE) +
  geom_point(color = "black") +
  scale_shape_manual(values = c(21, 22, 25)) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
ggsave("img/04Rl32N.png", width = 14, height = 7, units = "cm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

6.2 modèle initial, régression multiple

```r
regression_multiple <- lm(
   formula  = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length *
exp(Sepal.Width),
  data = iris
  )

car::Anova(regression_multiple, type = "3")
```

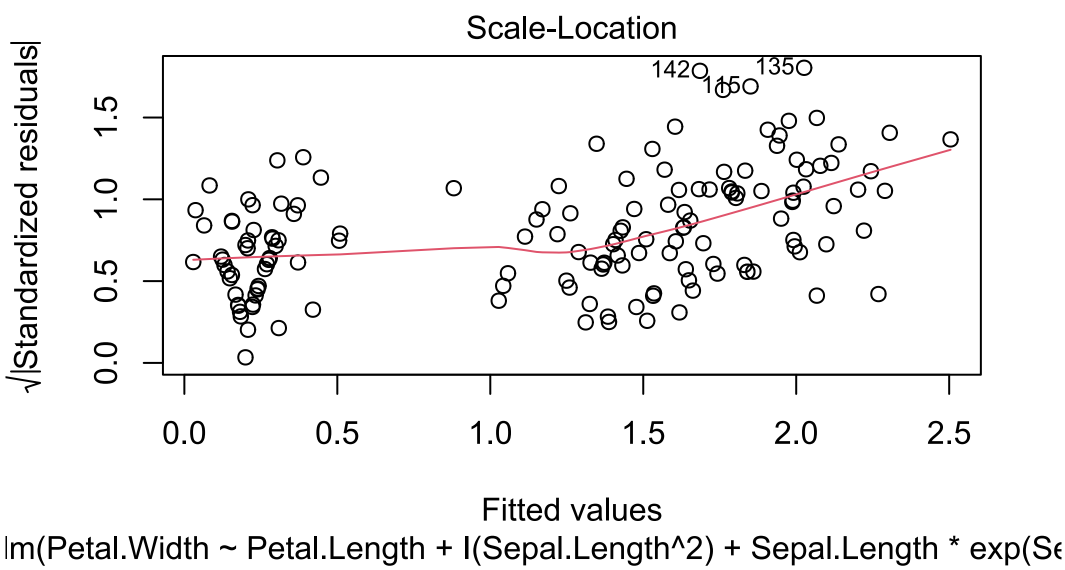
```
Anova Table (Type III tests)

Response: Petal.Width
                              Sum Sq  Df  F value  Pr(>F)
(Intercept)                   0.0654   1   1.7179 0.19205
Petal.Length                 12.4346   1 326.5651 < 2e-16 ***
I(Sepal.Length^2)             0.1266   1   3.3243 0.07034 .
Sepal.Length                  0.0413   1   1.0840 0.29954
exp(Sepal.Width)              0.0071   1   0.1874 0.66575
```

```
Sepal.Length:exp(Sepal.Width)  0.0342   1   0.8974 0.34506
Residuals                      5.4831 144
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
shapiro.test(regression_multiple$residuals)
```

```

	Shapiro-Wilk normality test

data:  regression_multiple$residuals
W = 0.98367, p-value = 0.07271
```

```r
plot(regression_multiple, which = 3)
```



deuxième modèle

```r
m2 <- lm(
   formula = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
exp(Sepal.Width),
  data = iris
  )
```

```r
car::Anova(m2, type = "3")
```

```
Anova Table (Type III tests)

Response: Petal.Width
                   Sum Sq  Df  F value    Pr(>F)
(Intercept)        0.0672   1   1.7656 0.1860137
Petal.Length      14.1860   1 372.8273 < 2.2e-16 ***
I(Sepal.Length^2)  0.0981   1   2.5776 0.1105631
Sepal.Length       0.0305   1   0.8019 0.3719993
exp(Sepal.Width)   0.4758   1  12.5049 0.0005454 ***
Residuals          5.5172 145
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(regression_multiple, m2)
```

```
Analysis of Variance Table

Model 1: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length *
    exp(Sepal.Width)
Model 2: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
    exp(Sepal.Width)
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    144 5.4831
2    145 5.5172 -1  -0.03417 0.8974 0.3451
```

```r
AIC(regression_multiple, m2)
```

```
                    df       AIC
regression_multiple  7 -56.66406
m2                   6 -57.73216
```

troisième modèle

```r
m3 <- lm(
  formula = Petal.Width ~ Petal.Length + I(Sepal.Length^2) + exp(Sepal.Width),
  data = iris
  )
```

```r
car::Anova(m3, type = "3")
```

```
Anova Table (Type III tests)

Response: Petal.Width
                   Sum Sq  Df F value    Pr(>F)
(Intercept)        0.9337   1  24.571 1.961e-06 ***
Petal.Length      17.9475   1 472.324 < 2.2e-16 ***
I(Sepal.Length^2)  0.6303   1  16.587 7.592e-05 ***
exp(Sepal.Width)   0.5502   1  14.479 0.0002079 ***
Residuals          5.5478 146
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(m3, m2)
```

```
Analysis of Variance Table

Model 1: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + exp(Sepal.Width)
Model 2: Petal.Width ~ Petal.Length + I(Sepal.Length^2) + Sepal.Length +
    exp(Sepal.Width)
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    146 5.5478
2    145 5.5172  1  0.030514 0.8019  0.372
```

```r
AIC(m3, m2)
```

```
   df       AIC
m3  5 -58.90486
m2  6 -57.73216
```

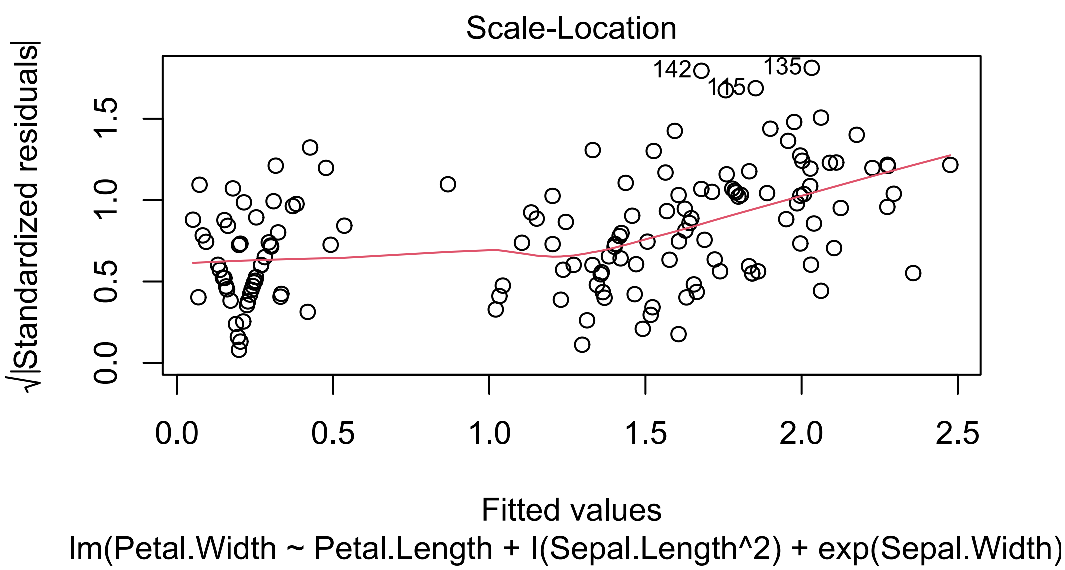
validation du modèle final

```r
shapiro.test(m3$residuals)
```

```

    Shapiro-Wilk normality test

```

```
data:  m3$residuals
W = 0.98388, p-value = 0.07691
```

```r
plot(m3, which = 3)
```



7 Regression non paramétrique univariée

7.1 dualité biais-variance

```r
ggplot(airquality) +
  aes(x = Temp, y = Ozone) +
  geom_point(color = "grey") +
  geom_smooth(method = "loess", linetype = "twodash", span = 0.2, color = "red",
se = FALSE) +
  geom_smooth(method = "loess", linetype = "dotdash", span = 2, color = "green",
se = FALSE) +
  geom_smooth(method = "loess", span = 0.75, color = "blue", se = FALSE) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

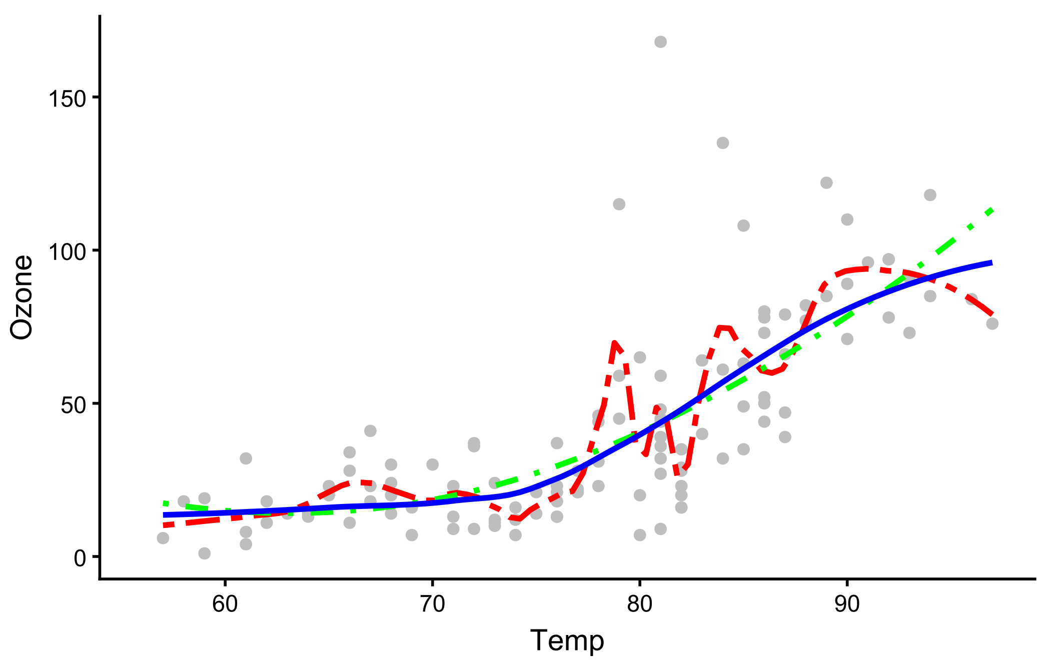
```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```




```
ggsave("img/04Rl39N.png", width = 14, height = 7, units = "cm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 37 rows containing missing values or values outside the scale range
(`geom_point()`).
```

7.2 Ozone ~ Temp

```
library(gam)
```

```
Warning: le package 'gam' a été compilé avec la version R 4.5.3
```

```
Le chargement a nécessité le package : splines
```

```
Le chargement a nécessité le package : foreach
```

```


Attachement du package : 'foreach'
```

```
Les objets suivants sont masqués depuis 'package:purrr':

    accumulate, when
```

```
Loaded gam 1.22-7
```

```
gam_uni <- gam(
  Ozone ~ lo(Temp),
  data = airquality
```

```r
)

summary(gam_uni)
```

```

Call: gam(formula = Ozone ~ lo(Temp), data = airquality)
Deviance Residuals:
    Min      1Q  Median      3Q     Max 
-34.999 -11.204  -3.733   8.843 124.001 

(Dispersion Parameter for gaussian family taken to be 487.0196)

    Null Deviance: 125143.1 on 115 degrees of freedom
Residual Deviance: 54285.01 on 111.4637 degrees of freedom
AIC: 1053.482 
37 observations effacées parce que manquantes

Number of Local Scoring Iterations: NA 

Anova for Parametric Effects
              Df Sum Sq Mean Sq F value    Pr(>F)    
lo(Temp)    1.00  61033   61033  125.32 < 2.2e-16 ***
Residuals 111.46  54285     487                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects
            Npar Df Npar F     Pr(F)    
(Intercept)                             
lo(Temp)        2.5  7.954 0.0001904 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
regression_exponentielle <- lm(Ozone ~ exp(Temp), data = airquality)
regression_polynomiale <- lm(Ozone ~ Temp + I(Temp^2), data = airquality)
AIC(gam_uni, regression_exponentielle, regression_polynomiale)
```

```
                         df      AIC
gam_uni                   3 1053.482
regression_exponentielle  3 1142.483
regression_polynomiale    4 1056.148
```

```
airquality |>
  rownames_to_column() |>
  full_join(
    predict(gam_uni) |>
      as.data.frame() |>
      rownames_to_column() |>
      rename(ozone_predit = "predict(gam_uni)")
  ) |>
  ggplot() +
  aes(x = Temp, y = Ozone) +
  geom_jitter() +
  geom_smooth(
    method = lm,
    formula = y ~ exp(x),
    data  = airquality |> filter(Temp < 86),
    linetype = "twodash",
    se = FALSE
    ) +
  geom_smooth(
    method = lm,
    formula = y ~ x + I(x^2),
    color = "green",
    se = FALSE
    ) +
  geom_smooth(
    aes(y = ozone_predit),
    linetype = "longdash",
    color = "red"
    ) +
  theme_classic()
```

```
Joining with `by = join_by(rowname)`
```

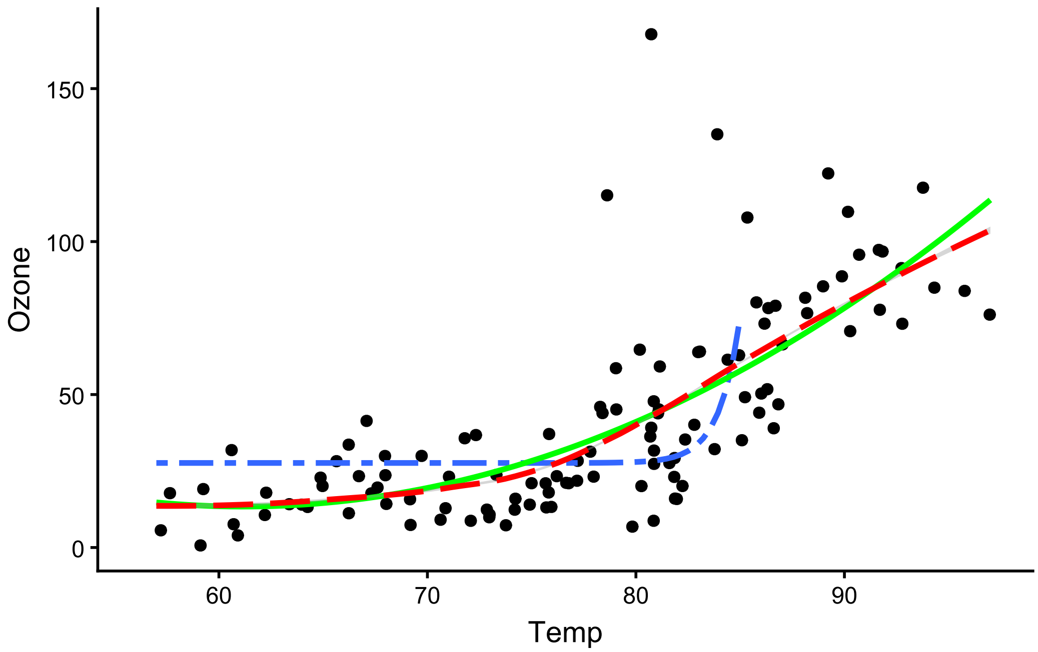
```
Warning: Removed 30 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```




```
ggsave("img/04Rl42N.png", width = 14, height = 7, units = "cm")
```

```
Warning: Removed 30 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale
range
(`geom_point()`).
```

7.3 regression_non_parametrique_multivariee

```
library(gam)

gam_initial <- gam(
  Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R) + lo(Day),
  data = airquality
)

summary(gam_initial)
```

```

Call: gam(formula = Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R) + lo(Day),
    data = airquality)
Deviance Residuals:
    Min      1Q  Median      3Q     Max
-43.074  -9.201  -2.155   7.836  65.410

(Dispersion Parameter for gaussian family taken to be 298.2742)

    Null Deviance: 121801.9 on 110 degrees of freedom
Residual Deviance: 28469.42 on 95.4471 degrees of freedom
AIC: 963.8333
42 observations effacées parce que manquantes

Number of Local Scoring Iterations: NA

Anova for Parametric Effects
                Df Sum Sq Mean Sq  F value    Pr(>F)
lo(Temp)     1.000  47237   47237 158.3680 < 2.2e-16 ***
lo(Wind)     1.000  11164   11164  37.4270 2.082e-08 ***
lo(Solar.R)  1.000   3724    3724  12.4863 0.0006339 ***
lo(Day)      1.000   1102    1102   3.6952 0.0575512 .
Residuals   95.447  28469     298
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects
            Npar Df  Npar F     Pr(F)
```

```
(Intercept)
lo(Temp)        2.5  7.0290  0.000564 ***
lo(Wind)        2.8 10.9651 4.782e-06 ***
lo(Solar.R)     2.6  1.5648  0.208397
lo(Day)         2.6  1.1102  0.344659
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

gam_2

```r
gam_2 <- gam(
  Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R),
  data = airquality
)

AIC(gam_initial, gam_2)
```

```
            df      AIC
gam_initial  6 963.8333
gam_2        5 963.6606
```

```r
summary(gam_2)
```

```

Call: gam(formula = Ozone ~ lo(Temp) + lo(Wind) + lo(Solar.R), data = airquality)
Deviance Residuals:
    Min      1Q  Median      3Q     Max
-46.417  -9.773  -3.465   8.943  71.975

(Dispersion Parameter for gaussian family taken to be 306.3018)

    Null Deviance: 121801.9 on 110 degrees of freedom
Residual Deviance: 30348.83 on 99.0815 degrees of freedom
AIC: 963.6606
42 observations effacées parce que manquantes

Number of Local Scoring Iterations: NA

Anova for Parametric Effects
                Df Sum Sq Mean Sq F value    Pr(>F)
lo(Temp)     1.000  48471   48471 158.246 < 2.2e-16 ***
```

```
lo(Wind)      1.000  11113   11113  36.282 2.914e-08 ***
lo(Solar.R)  1.000   3472    3472  11.335  0.001084 **
Residuals    99.081  30349     306
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Anova for Nonparametric Effects
            Npar Df Npar F     Pr(F)
(Intercept)
lo(Temp)        2.5 6.9452 0.0006017 ***
lo(Wind)        2.8 9.9053 1.386e-05 ***
lo(Solar.R)     2.6 2.3287 0.0885503 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

gam_3

```
gam_3 <- gam(
  Ozone ~ lo(Temp) + lo(Wind),
  data = airquality
)

AIC(gam_2, gam_3)
```

```
Warning in AIC.default(gam_2, gam_3): tous les modèles n'ont pas été ajustés
sur le même nombre d'observations
```

```
      df       AIC
gam_2  5  963.6606
gam_3  4 1019.0723
```

```
gam_3_bis <- gam(
  Ozone ~ lo(Temp) + lo(Wind),
  data = airquality |> filter(!is.na(Solar.R))
)

AIC(gam_2, gam_3_bis)
```

```
          df      AIC
gam_2      5 963.6606
gam_3_bis  4 972.6289
```